

Project Initialisation and Planning Phase

Date	02 July 2026
Team ID	xxxxxxx
Project Title	OptiCrop:Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Opti Crop system is a Machine Learning-based crop recommendation application that recommends suitable crops based on soil nutrients and environmental conditions. Using Python, Flask, and Machine Learning, the system helps farmers improve productivity and make data-driven farming decisions.

Project Overview	
Objective	To develop a Machine Learning-based system that recommends the most suitable crop using soil nutrients and environmental conditions.
Scope	The project includes data preprocessing, model training, prediction and deployment of a Flask web application for real-time crop Recommendation.
Problem Statement	
Description	Selecting suitable crops based on traditional methods may reduce productivity. This project provides intelligent crop recommendations using Machine Learning. Improves crop yield, supports sustainable farming, and helps farmers make informed agricultural decisions.
Impact	
Proposed Solution	
Approach	The system analyses N, P, K, temperature, humidity, pH, and rainfall using a trained Machine Learning model integrated with a Flask web application.

Key Features	-Machine Learning-based crop recommendation -Real-time prediction using soil and climate data -User-friendly Flask web application -Fast and accurate crop recommendations -Supports sustainable and data-driven farming
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	- Easy integration with agricultural datasets - Optimised resource utilisation for better crop yield
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / Apple M1
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	10 GB Available Storage
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, Matplotlib, Pickle
Development Environment	IDE	Jupyter Notebook, VScode
Data		
Data	Source, size, format	Crop Recommendation dataset (CSV)